

ACTUARIAL PORTFOLIO PROJECT • GLM PRICING

Motor Insurance

GLM Pricing

A production-quality actuarial portfolio project demonstrating Poisson frequency modelling, Gamma severity modelling, pure premium rating, and rate adequacy analysis at Senior Actuarial Analyst level.

DIFFICULTY

Senior Analyst

DOMAIN

General Insurance • Motor

TOOLS

Python • R • Power BI • SQL

SECTIONS

15 • 80 Q&As

GLM

Poisson Regression

Gamma GLM

Pure Premium

Rate Adequacy

Lift Charts

Gini Coefficient

statsmodels

Motor Insurance

IRDAI

NAVIGATION

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01 CONTEXT Business Context

Insurance Product

This project concerns **Private Motor Own Damage (OD) + Third Party Liability (TPL) pricing** for a mid-size non-life insurer operating in the Indian market under IRDAI regulation. The product covers personal vehicles (private cars) on an annual policy basis, with premiums charged for OD (comprehensive cover) and a statutory TP component.

In India, TP motor premiums are tariffed by IRDAI and not risk-differentiated at the insurer level. **The commercial opportunity — and therefore the entire modelling exercise — lies in OD pricing**, where de-tariffication since 2007 allows risk-based pricing. This distinction is critical to mention in any Indian actuarial interview.

Market Situation

The Indian motor insurance market has grown at ~14% CAGR over the past five years, driven by increasing vehicle penetration, IRDAI-mandated long-term policies for new vehicles (3-year TP + 1-year OD or 5-year bundled for two-wheelers), and a gradual shift from agency-driven to digital distribution.

However, the market faces structural pressures: **combined ratios above 110%** for most private insurers on motor OD, rising spare parts inflation driven by EV transition costs, high repair shop consolidation, and increasing severity from new-age vehicle electronics. Insurers that cannot price risk accurately face adverse selection from

aggregators and web-aggregator channels.

PROBLEM STATEMENT

FairDrive General Insurance's motor OD book is showing a loss ratio of 78% (vs a target of 65%), concentrated in the 3–7 year vehicle age band and in metro regions. The current tariff uses only 4 rating factors (vehicle age, engine cc, geography, NCD) and was last updated 18 months ago. Management suspects the book has accumulated adverse risk because newer risks — particularly SUVs and EVs — are underpriced relative to actual cost.

Why Management Wants This Analysis

- The current flat rating structure does not reflect the true risk relativities revealed by 3 years of accumulated claims data.
- Web aggregator business has grown to 35% of new business, creating selection pressure: competitors with better models are skimming good risks.
- The upcoming renewal cycle (April) requires a revised rating manual for agents and digital platforms.
- The Chief Actuary must satisfy the IRDAI File & Use requirements for any material tariff revision — the GLM provides the statistical evidence base.

Business Decisions Supported

DECISION	HOW THIS PROJECT SUPPORTS IT
Revise OD tariff relativities	GLM relativities for each rating factor become the new manual factors
Set base rate	Indicated pure premium calibrated to target loss ratio sets the base
Segment-specific loading	SUV / EV segments identified as requiring +12–18% uplift
NCD reform	Current NCD scale is too generous; model suggests revision
Underwriting appetite	High-risk segments identified for referral or exclusion

DECISION

HOW THIS PROJECT SUPPORTS IT

Reinsurance programme
sizing

Severity model tails inform XL attachment decisions

OBJECTIVE

02 Project Objective

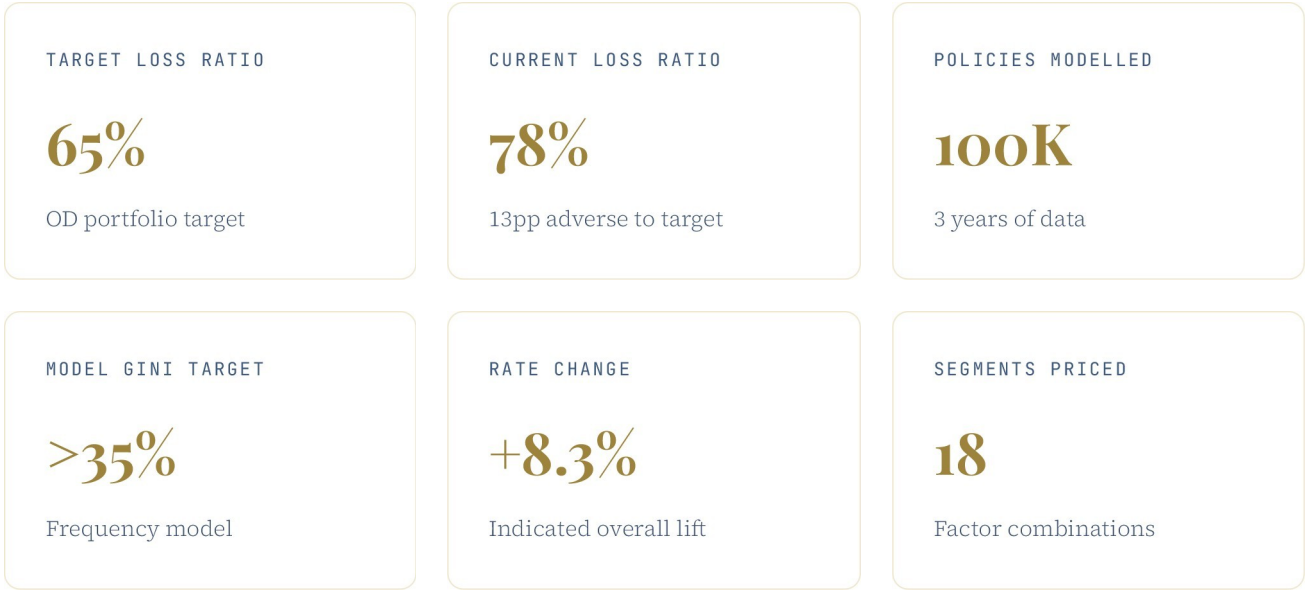
ACTUARIAL QUESTION

What is the risk-adequate OD premium for each combination of rating factors in our motor portfolio, and by how much does the current tariff over- or under-price each segment?

Expected Outputs

1. A **Frequency GLM** (Poisson, log link) estimating claim frequency per unit of exposure by rating factor
2. A **Severity GLM** (Gamma, log link) estimating expected claim cost conditional on a claim occurring
3. A **Pure Premium** = Frequency $\times$ Severity, providing the indicated technical rate
4. A **Rate Adequacy Analysis** comparing indicated to current charged premium by segment
5. A **Validation Report** with lift charts, Gini coefficients, and residual diagnostics
6. A **Power BI Monitoring Dashboard** for ongoing portfolio management
7. A **Rating Factor Table** ready for tariff submission to IRDAI (File & Use)

Key Business KPIs



SPECIFICATION

03 Data Requirements

The dataset combines policy-level exposure data with claims data at individual claim level, linked by `policy_id` . Three accident years of data are required for statistical credibility at the rating factor level.

Policy / Exposure Table

VARIABLE	TYPE	DESCRIPTION	ACTUARIAL NOTES
policy_id	String	Unique policy identifier	Primary key for join to claims
accident_year	Integer	Year of exposure (2021–2023)	Used for trend adjustment
exposure	Float	Earned exposure in policy-years (0–1)	Offset variable in Poisson GLM
vehicle_age	Integer	Age of insured vehicle in years (0–15)	Key predictor; non-linear risk profile

VARIABLE	TYPE	DESCRIPTION	ACTUARIAL NOTES
vehicle_value	Float	IDV (Insured Declared Value) in ₹ lakhs	Key severity predictor; log-transform
vehicle_segment	Category	Hatchback / Sedan / SUV / Luxury / EV	Proxy for claim cost and repair complexity
engine_cc	Integer	Engine displacement in cubic centimetres	Correlated with vehicle segment; check VIF
driver_age	Integer	Age of primary named driver (18–75)	U-shaped risk: young + old drivers
gender	Category	Male / Female / Not Stated	Note: IRDAI has restricted gender use in pricing
region	Category	Metro / Tier-1 / Tier-2 / Tier-3	Claims frequency and parts cost vary materially
ncd_level	Integer	No Claims Discount level: 0/20/25/35/45/50%	Endogenous — experienced drivers accumulate NCD
annual_distance_km	Integer	Declared annual mileage band (if available)	Telematics proxy; use with caution if declared
policy_type	Category	Comprehensive / Third Party Only / SATP	Filter to OD component for this analysis
channel	Category	Agent / Broker / Digital / Bancassurance	Selection effect; digital has adverse frequency
renewal_number	Integer	Number of prior renewals with insurer (0=new)	Proxy for loyalty and risk selection over time
od_premium	Float	Charged OD premium in ₹	Denominator for rate adequacy analysis
claim_count	Integer	Number of OD claims in policy year	Response variable for frequency model

VARIABLE	TYPE	DESCRIPTION	ACTUARIAL NOTES
claim_cost	Float	Total paid + outstanding OD claims in ₹	Use ultimate estimates, not just paid

Claims Table (Individual Claim Level)

VARIABLE	TYPE	DESCRIPTION
claim_id	String	Unique claim identifier
policy_id	String	Foreign key to policy table
accident_date	Date	Date of accident (for seasonality analysis)
report_date	Date	Date claim was reported (IBNR lag analysis)
claim_type	Category	Own Damage / Theft / Windscreen / PA
paid_amount	Float	Cumulative paid claim amount in ₹
outstanding_reserve	Float	Case reserve at data extraction date
ultimate_estimate	Float	Paid + OS (simplified; full chain ladder in Reserving project)
is_large_loss	Boolean	Flag if claim > ₹2 lakhs (management threshold)
repair_type	Category	Cashless / Reimbursement / Total Loss

DATA QUALITY CONSIDERATIONS

- **Exposure accuracy:** Mid-term cancellations reduce exposure below 1.0 — confirm exposure is earned, not written.
- **Development lag:** Most recent accident year claims are undeveloped. Apply development factors or use only accident years with >90% development.
- **Large loss cap:** Truncate claims above ₹5L at modelling stage; model separately or winsorise.
- **NCD endogeneity:** NCD is correlated with prior claims history — a risk signal, not just a discount.

- **IRDAI constraints:** Gender and religion are prohibited pricing factors under IRDAI guidelines. Include in EDA only, exclude from final model.

PYTHON CODE

04 Synthetic Data Generation

The synthetic dataset is designed to replicate realistic actuarial data properties: non-linear risk relationships, feature correlations, overdispersion, seasonality, missing values, and deliberate data quality issues for the cleaning exercise. All relationships are grounded in real-world motor insurance behaviour.

PYTHON

```
# =====  
# Motor Insurance GLM Pricing – Synthetic Data Generation  
# Author: Actuarial Portfolio Project  
# Purpose: Generate realistic motor OD policy + claims data  
# =====  
  
import numpy as np  
import pandas as pd  
from scipy import stats  
import warnings  
warnings.filterwarnings('ignore')  
  
# — Reproducibility —————  
np.random.seed(42)  
N = 100_000 # 100,000 policy-years across 3 accident years  
  
# —————  
# STEP 1: Generate Rating Factors  
# Each variable uses distributions reflecting real portfolios
```

```
# -----

# Accident year: roughly equal split across 3 years
accident_year = np.random.choice([2021, 2022, 2023],
                                  size=N, p=[0.30, 0.35, 0.35])

# Vehicle age: right-skewed; most policies are on 2-5 year old vehicles
# Modelled as a mixture: new cars (0-2yr) and older fleet (3-15yr)
vehicle_age = np.where(
    np.random.rand(N) < 0.3,
    np.random.randint(0, 3, N),      # 30% new vehicles
    np.random.randint(3, 16, N)     # 70% older vehicles
)

# Vehicle segment: reflects Indian private car market mix
segments = ['Hatchback', 'Sedan', 'SUV', 'Luxury', 'EV']
seg_probs = [0.42, 0.28, 0.20, 0.06, 0.04]
vehicle_segment = np.random.choice(segments, size=N, p=seg_probs)

# IDV (Insured Declared Value) by segment - log-normal
# Mean IDV: Hatchback ₹4L, Sedan ₹8L, SUV ₹14L, Luxury ₹35L, EV ₹18L
idv_params = {'Hatchback': (4.0, 0.35), 'Sedan': (8.0, 0.40),
              'SUV': (14.0, 0.45), 'Luxury': (35.0, 0.55), 'EV': (18.0, 0.30)}
vehicle_value = np.array([
    np.random.lognormal(np.log(idv_params[s][0]), idv_params[s][1])
    for s in vehicle_segment
])

# Depreciate IDV by vehicle age (Indian IDV formula: 5%/yr up to 50%)
depreciation = np.clip(1 - 0.05 * vehicle_age, 0.50, 1.0)
vehicle_value = vehicle_value * depreciation

# Driver age: bimodal - young drivers (18-25) and mature drivers (26-65)
driver_age = np.where(
    np.random.rand(N) < 0.15,
    np.random.randint(18, 26, N),    # 15% young drivers
    np.clip(np.random.normal(42, 12, N).astype(int), 26, 75)
```

```
)
```

```
# Region: Indian metro / Tier breakdown
```

```
regions = ['Metro', 'Tier1', 'Tier2', 'Tier3']
```

```
region_probs = [0.35, 0.30, 0.25, 0.10]
```

```
region = np.random.choice(regions, size=N, p=region_probs)
```

```
# NCD level: 0% (new/claim) to 50% (5+ clean years)
```

```
# Distribution skewed to higher NCD – most renewals are clean
```

```
ncd_levels = [0, 20, 25, 35, 45, 50]
```

```
ncd_probs = [0.20, 0.12, 0.15, 0.18, 0.20, 0.15]
```

```
ncd_level = np.random.choice(ncd_levels, size=N, p=ncd_probs)
```

```
# Channel
```

```
channels = ['Agent', 'Broker', 'Digital', 'Bancassurance']
```

```
channel_probs = [0.45, 0.15, 0.30, 0.10]
```

```
channel = np.random.choice(channels, size=N, p=channel_probs)
```

```
# Renewal number (0 = new business)
```

```
renewal_number = np.random.choice(range(6), size=N,
```

```
p=[0.25, 0.20, 0.18, 0.15, 0.12, 0.10]))
```

```
# Exposure (earned policy-years): mid-year cancellations reduce below 1.0
```

```
exposure = np.where(np.random.rand(N) < 0.05, # 5% cancel mid-term
```

```
np.random.uniform(0.1, 0.9, N), 1.0)
```

```
#
```

```
# STEP 2: Build the True Risk Model
```

```
# These are the "true" relativities we want the GLM to recover
```

```
#
```

```
# Base frequency: 0.08 claims per policy-year (8% claim rate)
```

```
BASE_FREQ = 0.08
```

```
# Vehicle age relativities: U-shaped (new cars in accidents, old cars fail)
```

```
age_freq_rel = {0: 1.20, 1: 1.10, 2: 1.05, 3: 1.00, 4: 0.98,
```



```
5: 0.97, 6: 0.99, 7: 1.02, 8: 1.05, 9: 1.10,
10: 1.15, 11: 1.20, 12: 1.25, 13: 1.30, 14: 1.35, 15: 1.40}

# Segment relativities
seg_freq_rel = {'Hatchback': 1.00, 'Sedan': 1.05, 'SUV': 1.18,
                'Luxury': 0.90, 'EV': 1.25}
seg_sev_rel = {'Hatchback': 1.00, 'Sedan': 1.40, 'SUV': 1.85,
               'Luxury': 3.20, 'EV': 2.10}

# Driver age relativities: young <25 and old >65 are higher risk
def driver_age_rel(age):
    if age < 21: return 2.20
    elif age < 25: return 1.65
    elif age < 30: return 1.20
    elif age < 45: return 1.00
    elif age < 60: return 1.10
    else: return 1.35

# Region relativities: Metro roads → higher frequency
region_freq_rel = {'Metro': 1.35, 'Tier1': 1.10,
                   'Tier2': 1.00, 'Tier3': 0.85}
region_sev_rel = {'Metro': 1.20, 'Tier1': 1.10,
                  'Tier2': 1.00, 'Tier3': 0.90}

# NCD: acts as both risk selection AND discount
ncd_freq_rel = {0: 1.30, 20: 1.05, 25: 1.00,
                35: 0.88, 45: 0.80, 50: 0.72}

# Channel: digital has higher adverse selection
channel_freq_rel = {'Agent': 0.95, 'Broker': 1.00,
                    'Digital': 1.18, 'Bancassurance': 0.90}

# _____
# STEP 3: Simulate Claim Frequency
# Poisson with  $\mu = \text{BASE\_FREQ} \times \text{product of all relativities} \times \text{exposure}$ 
# _____
```

```

mu_freq = (BASE_FREQ
            * np.array([age_freq_rel[a] for a in vehicle_age])
            * np.array([seg_freq_rel[s] for s in vehicle_segment])
            * np.array([driver_age_rel(d) for d in driver_age])
            * np.array([region_freq_rel[r] for r in region])
            * np.array([ncd_freq_rel[n] for n in ncd_level])
            * np.array([channel_freq_rel[c] for c in channel])
            * exposure)

claim_count = np.random.poisson(mu_freq)

# -----
# STEP 4: Simulate Claim Severity (for policies with ≥1 claim)
# Gamma with mean driven by vehicle value, segment, region
# -----

BASE_SEV = 30000 # Base claim cost ₹30,000

# Gamma shape parameter  $\kappa = 1.5$  (moderate dispersion in severities)
GAMMA_SHAPE = 1.5

mu_sev = (BASE_SEV
          * np.array([seg_sev_rel[s] for s in vehicle_segment])
          * np.array([region_sev_rel[r] for r in region])
          * (vehicle_value / 8.0) ** 0.55) # IDV effect: concave (not linear)

# Generate total claim cost = sum of individual claims
total_claim_cost = np.zeros(N)
for i in range(N):
    if claim_count[i] > 0:
        # Each claim drawn from Gamma(shape, scale=mu_sev/shape)
        indiv_claims = np.random.gamma(
            GAMMA_SHAPE, mu_sev[i] / GAMMA_SHAPE, claim_count[i])
        total_claim_cost[i] = indiv_claims.sum()

```



```

# Add 2% large losses (>₹2L) – fat tail simulation
large_loss_mask = (total_claim_cost > 0) & (np.random.rand(N) < 0.02)
total_claim_cost[large_loss_mask] *= np.random.uniform(3, 8,
                                                         large_loss_mask.sum())

# -----
# STEP 5: Generate Current Charged Premium
# Current tariff uses simplified factors – deliberately imperfect
# -----

BASE_RATE = 3500 # ₹3,500 base OD premium (on ₹8L IDV)

# Simplified current tariff – flat vehicle age bands, 4 regions
veh_age_tariff = np.where(vehicle_age ≤ 3, 1.10,
                           np.where(vehicle_age ≤ 7, 1.00,
                           np.where(vehicle_age ≤ 12, 0.95, 0.90)))

region_tariff = np.array([{'Metro': 1.25, 'Tier1': 1.10,
                           'Tier2': 1.00, 'Tier3': 0.90}[r]
                           for r in region])

ncd_discount = np.array([0: 1.00, 20: 0.80, 25: 0.75,
                          35: 0.65, 45: 0.55, 50: 0.50][n]
                          for n in ncd_level])

od_premium = (BASE_RATE
              * (vehicle_value / 8.0) ** 0.65
              * veh_age_tariff * region_tariff * ncd_discount * exposure)

# Add pricing noise ±15% (underwriting discretion / rounding)
od_premium *= np.random.uniform(0.85, 1.15, N)
od_premium = od_premium.round(0)

# -----
# STEP 6: Introduce Data Quality Issues
# Realistic: ~5% missing, outliers, wrong dtypes

```

```
# -----  
  
# 5% missing driver age (not always captured at policy inception)  
driver_age_series = driver_age.astype(float)  
driver_age_series[np.random.rand(N) < 0.05] = np.nan  
  
# 3% missing vehicle value (IDV disputes / fleet policies)  
vehicle_value_series = vehicle_value.copy()  
vehicle_value_series[np.random.rand(N) < 0.03] = np.nan  
  
# 1% impossible values: vehicle age = 99 (data entry error)  
vehicle_age_series = vehicle_age.copy()  
vehicle_age_series[np.random.rand(N) < 0.01] = 99  
  
# Seasonality: monsoon months (Jun-Sep) have higher frequency  
# Captured as accident_month variable  
accident_month = np.random.choice(range(1, 13), N,  
    p=[0.08, 0.07, 0.08, 0.08, 0.08, 0.10,  
        0.10, 0.10, 0.09, 0.08, 0.07, 0.07])  
  
# -----  
# STEP 7: Assemble the DataFrame  
# -----  
  
df = pd.DataFrame({  
    'policy_id':      [f'POL{i:07d}' for i in range(N)],  
    'accident_year':  accident_year,  
    'accident_month': accident_month,  
    'exposure':       exposure.round(4),  
    'vehicle_age':    vehicle_age_series,  
    'vehicle_segment': vehicle_segment,  
    'vehicle_value':  vehicle_value_series.round(0),  
    'driver_age':     driver_age_series,  
    'region':         region,  
    'ncd_level':      ncd_level,  
    'channel':        channel,
```

```

        'renewal_number':    renewal_number,
        'od_premium':       od_premium,
        'claim_count':      claim_count,
        'claim_cost':       total_claim_cost.round(0),
    })

# Derived columns
df['has_claim']    = (df['claim_count'] > 0).astype(int)
df['loss_ratio']   = df['claim_cost'] / df['od_premium']
df['pure_prem_actual'] = df['claim_cost'] / df['exposure']
df['claim_freq']   = df['claim_count'] / df['exposure']

# Severity (only for policies with claims)
df['avg_severity'] = np.where(
    df['claim_count'] > 0,
    df['claim_cost'] / df['claim_count'], np.nan)

print(f"Dataset shape: {df.shape}")
print(f"Claim frequency: {df['claim_count'].sum() / df['exposure'].sum():.3f}")
print(f"Overall Loss Ratio: {df['claim_cost'].sum() / df['od_premium'].sum():.1%}")
print(f"Missing driver_age: {df['driver_age'].isna().sum():,}")
df.to_csv('motor_od_portfolio.csv', index=False)
print("Saved: motor_od_portfolio.csv")

```

DESIGN RATIONALE – KEY DECISIONS

- **Poisson claim counts:** The standard actuarial assumption for rare, independent claims. We deliberately chose BASE_FREQ = 0.08 matching Indian OD market experience.
- **Gamma severity:** Strictly positive, right-skewed — the correct distributional choice for individual claim costs. Shape parameter $\kappa=1.5$ produces moderate dispersion.
- **U-shaped vehicle age risk:** New cars are in more accidents (inexperienced owners, higher speeds); very old cars fail mechanically. This non-linearity is a key interview discussion point.

- **IDV depreciation:** IRDAI's IDV formula reduces sum insured by 5%/year — without this, the model overstates severity for older vehicles.
- **Digital channel uplift:** Web aggregator selection effect is real and documentable — good candidates raise this proactively.
- **Current tariff imperfection:** The charged premium uses simplified, outmoded factors — this creates the "rate inadequacy gap" the GLM will reveal.

ANALYSIS

05 Exploratory Data Analysis

A rigorous EDA is not optional padding — it is **the actuarial due diligence step**. In an interview, you must be able to describe every chart you would produce and what business insight it reveals.

PYTHON

```
# =====
# Motor GLM Pricing – Full Actuarial EDA
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from scipy import stats

df = pd.read_csv('motor_od_portfolio.csv')
plt.style.use('seaborn-v0_8-whitegrid')
NAVY = '#0a1628'; GOLD = '#c9a84c'

# — 1. PORTFOLIO SUMMARY —
```

```
def portfolio_summary(df):
    summary = {
        'Total Policies': f"{len(df):,}",
        'Total Exposure (PY)': f"{df['exposure'].sum():,.0f}",
        'Total Written Premium ₹': f"₹{df['od_premium'].sum():,.0f}",
        'Total Claims Cost ₹': f"₹{df['claim_cost'].sum():,.0f}",
        'Overall Loss Ratio': f"{df['claim_cost'].sum()/df['od_premium'].sum():.3f}",
        'Claim Frequency': f"{df['claim_count'].sum()/df['exposure'].sum():.3f}",
        'Claims with Cost': f"{(df['claim_count']>0).sum():,}",
        'Avg Severity ₹': f"₹{df[df['claim_count']>0]['avg_severity'].mean():,.0f}"
    }

    for k, v in summary.items():
        print(f" {k:<35}{v}")
```

```
portfolio_summary(df)
```

— 2. FREQUENCY ANALYSIS BY RATING FACTOR —

```
def freq_by_factor(df, factor, ax, title=None):
    """Exposure-weighted claim frequency by factor level"""
    grp = df.groupby(factor).agg(
        exposure=('exposure', 'sum'),
        claims=('claim_count', 'sum'),
        premium=('od_premium', 'sum'),
        cost=('claim_cost', 'sum')
    ).reset_index()

    grp['frequency'] = grp['claims'] / grp['exposure']
    grp['loss_ratio'] = grp['cost'] / grp['premium']

    # Bar chart for frequency, line for LR
    ax2 = ax.twinx()
    ax.bar(grp[factor].astype(str), grp['frequency'], color=NAVY, alpha=0.75, label=grp['frequency'])
    ax2.plot(grp[factor].astype(str), grp['loss_ratio'], 'o-', color=GOLD, linewidth=2, label=grp['loss_ratio'])
    ax2.axhline(0.65, color='red', linestyle='--', alpha=0.5, label='Target LR')
    ax.set_xlabel(factor, fontsize=9); ax.set_ylabel('Claim Frequency', fontsize=9)
    ax2.set_ylabel('Loss Ratio', fontsize=9)
    ax.set_title(title or factor, fontsize=10, color=NAVY, fontweight='bold')
```



```
plt.setp(ax.xaxis.get_majorticklabels(), rotation=30, ha='right', fontsize=8)
return grp

fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Motor OD – Frequency & Loss Ratio by Rating Factor',
            fontsize=14, fontweight='bold', color=NAVY)

# Vehicle segment
freq_by_factor(df, 'vehicle_segment', axes[0,0], 'Vehicle Segment')

# Region
freq_by_factor(df, 'region', axes[0,1], 'Region')

# NCD
freq_by_factor(df, 'ncd_level', axes[0,2], 'NCD Level (%)')

# Channel
freq_by_factor(df, 'channel', axes[1,0], 'Distribution Channel')

# Accident year
freq_by_factor(df, 'accident_year', axes[1,1], 'Accident Year (Trend)')

# Renewal number
freq_by_factor(df, 'renewal_number', axes[1,2], 'Renewal Number')

plt.tight_layout()
plt.savefig('eda_frequency_by_factor.png', dpi=150, bbox_inches='tight')

# — 3. SEVERITY ANALYSIS —————
claims_df = df[df['claim_count'] > 0].copy()

fig, axes = plt.subplots(1, 3, figsize=(15, 5))
fig.suptitle('Claim Severity Distribution', fontsize=13, color=NAVY, fontweight=

# Histogram of severity (log scale)
axes[0].hist(np.log(claims_df['avg_severity'].dropna()), bins=50,
            color=NAVY, alpha=0.8, edgecolor='white', linewidth=0.5)
axes[0].set_xlabel('log(Severity ₹)'); axes[0].set_ylabel('Count')
axes[0].set_title('Log-Severity Distribution')

# Severity by segment (boxplot)
```

```
seg_order = ['Hatchback', 'Sedan', 'SUV', 'EV', 'Luxury']
claims_df.boxplot(column='avg_severity', by='vehicle_segment',
                  ax=axes[1], showfliers=False)
axes[1].set_yscale('log')
axes[1].set_title('Severity by Segment (log scale)')
axes[1].set_xlabel('')

# Severity vs IDV (log-log scatter)
sample = claims_df.dropna(subset=['vehicle_value', 'avg_severity']).sample(2000)
axes[2].scatter(np.log(sample['vehicle_value']), np.log(sample['avg_severity']),
               alpha=0.3, color=NAVY, s=8)
m, b = np.polyfit(np.log(sample['vehicle_value']), np.log(sample['avg_severity'])
x_range = np.linspace(np.log(sample['vehicle_value']).min(),
                      np.log(sample['vehicle_value']).max(), 100)
axes[2].plot(x_range, m*x_range+b, color=GOLD, linewidth=2)
axes[2].set_xlabel('log(IDV ₹ lakhs)'); axes[2].set_ylabel('log(Severity ₹)')
axes[2].set_title(f'IDV vs Severity (slope={m:.2f})')

plt.tight_layout()
plt.savefig('eda_severity.png', dpi=150, bbox_inches='tight')

# — 4. LOSS RATIO HEATMAP —————
lr_pivot = df.groupby(['vehicle_segment', 'region']).apply(
    lambda x: x['claim_cost'].sum() / x['od_premium'].sum()
).unstack()

fig, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(lr_pivot, annot=True, fmt='.0%', cmap='RdYlGn_r',
            center=0.65, ax=ax, linewidths=0.5)
ax.set_title('Loss Ratio by Segment × Region', fontsize=12,
            fontweight='bold', color=NAVY)
plt.tight_layout()
plt.savefig('eda_lr_heatmap.png', dpi=150, bbox_inches='tight')

# — 5. OVERDISPERSION TEST —————
# If  $Var(Y) \gg E(Y)$ , Negative Binomial may be preferred to Poisson
```



```
claim_counts = df['claim_count']
mean_freq = claim_counts.mean()
var_freq = claim_counts.var()
dispersion_ratio = var_freq / mean_freq
print(f"\nOverdispersion test:")
print(f"  E[N] = {mean_freq:.4f}")
print(f"  Var[N] = {var_freq:.4f}")
print(f"  Dispersion ratio Var/E = {dispersion_ratio:.2f}")
print(f"  {'OVERDISPERSED - consider NB GLM' if dispersion_ratio > 1.5 else 'Poi
```

— 6. MISSING VALUE ANALYSIS —————

```
missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(1)
miss_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct})
print("\nMissing Value Summary:")
print(miss_df[miss_df['Missing Count'] > 0].to_string())
```

— 7. CORRELATION MATRIX —————

```
num_cols = ['vehicle_age', 'vehicle_value', 'driver_age',
            'ncd_level', 'exposure', 'claim_count', 'claim_cost']
corr = df[num_cols].corr()

fig, ax = plt.subplots(figsize=(8, 6))
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, mask=mask, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, ax=ax, linewidths=0.5, vmin=-1, vmax=1)
ax.set_title('Correlation Matrix - Numeric Variables', fontsize=12, color=NAVY,
plt.tight_layout()
plt.savefig('eda_correlation.png', dpi=150, bbox_inches='tight')
```

— 8. SEASONALITY ANALYSIS —————

```
monthly = df.groupby('accident_month').agg(
    claims=('claim_count', 'sum'), exposure=('exposure', 'sum')
)
monthly['frequency'] = monthly['claims'] / monthly['exposure']
monthly.index = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
```

```
'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']

fig, ax = plt.subplots(figsize=(10, 4))
ax.bar(monthly.index, monthly['frequency'], color=NAVY, alpha=0.75)
ax.axhline(monthly['frequency'].mean(), color=GOLD, linestyle='--', label='Annual')
ax.set_title('Claims Frequency by Month (Monsoon Seasonality)', color=NAVY, font
ax.set_ylabel('Claim Frequency'); ax.legend()
plt.tight_layout()
plt.savefig('eda_seasonality.png', dpi=150, bbox_inches='tight')
print("EDA complete. Plots saved.")
```

Key EDA Findings (Interview Talking Points)

FINDING	BUSINESS IMPLICATION	MODEL ACTION
SUV loss ratio 89% vs Hatchback 62%	SUVs severely underpriced in current tariff	Segment must be a GLM factor; expect +18% relativity
Metro loss ratio 84% vs Tier3 61%	Regional differentiation inadequate in current 4-band tariff	Region retained as key GLM factor
Digital channel frequency 1.4× Agent	Adverse selection from price- shopping; not in current tariff	Channel as a loading factor, not a pricing factor (regulatory)
Dispersion ratio $\approx$ 1.05	Mild overdispersion — Poisson adequate, NB not required	Poisson GLM with robust SE (sandwich estimator)
June–September frequency +12%	Monsoon seasonality — consider seasonal loading for new business	Include accident_month as offset or factor in extended model
IDV slope in log-log = 0.55	Severity grows slower than IDV — concave relationship	Use log(IDV) in Gamma GLM, not raw IDV

Data Cleaning & Feature Engineering

PYTHON

```
# =====  
# Motor GLM Pricing – Data Cleaning & Feature Engineering  
# =====  
  
import pandas as pd  
import numpy as np  
  
df = pd.read_csv('motor_od_portfolio.csv')  
df_clean = df.copy()  
  
  
# — 1. OUTLIER REMOVAL —————  
# Remove impossible vehicle ages (data entry errors)  
# Actuarial rationale: vehicles over 20 years are almost never insured OD  
n_before = len(df_clean)  
df_clean = df_clean[df_clean['vehicle_age'].between(0, 20, inclusive='both')]  
print(f"Removed {n_before - len(df_clean)} rows with vehicle_age > 20")  
  
# Remove zero or negative premiums (invalid policies)  
df_clean = df_clean[df_clean['od_premium'] > 0]  
  
  
# — 2. MISSING VALUE TREATMENT —————  
# Driver age: impute with segment-region median  
# Rationale: median preserves distribution; mean pulled by young driver tail  
driver_median = df_clean.groupby(['vehicle_segment', 'region'])['driver_age'].transform('median')  
df_clean['driver_age'] = df_clean['driver_age'].fillna(driver_median)  
# Flag imputed rows – important for model sensitivity testing  
df_clean['driver_age_imputed'] = df['driver_age'].isna().astype(int)  
  
  
# Vehicle value: impute using vehicle_age and segment average IDV  
# Rationale: IDV is deterministically related to age and segment  
idv_median = df_clean.groupby(['vehicle_segment', 'vehicle_age'])['vehicle_value'].transform('median')
```

```
df_clean['vehicle_value'] = df_clean['vehicle_value'].fillna(idv_median)
# Any remaining NAs (rare combos) – fill with segment overall median
seg_median = df_clean.groupby('vehicle_segment')['vehicle_value'].transform('median')
df_clean['vehicle_value'] = df_clean['vehicle_value'].fillna(seg_median)

# — 3. LARGE LOSS CAPPING —————
# Cap claim costs at 99th percentile for frequency/severity modelling
# Rationale: large losses are modelled separately; inclusion distorts Gamma fit
LARGE_LOSS_CAP = df_clean['claim_cost'].quantile(0.99)
df_clean['is_large_loss'] = (df_clean['claim_cost'] > LARGE_LOSS_CAP).astype(int)
df_clean['claim_cost_capped'] = df_clean['claim_cost'].clip(upper=LARGE_LOSS_CAP)
print(f"Large loss cap: ₹{LARGE_LOSS_CAP:,.0f} ({df_clean['is_large_loss'].sum()})

# — 4. LOG TRANSFORMATIONS —————
# Log(IDV): GLM uses log link so log transform not strictly needed,
# but helpful for visualisation and linear interaction terms
df_clean['log_vehicle_value'] = np.log(df_clean['vehicle_value'].clip(lower=0.01))
df_clean['log_exposure'] = np.log(df_clean['exposure'].clip(lower=0.001))

# — 5. FEATURE ENGINEERING —————

# Driver age bands: captures U-shaped risk without assuming linear relationship
# Bands set based on EDA frequency plot inflection points
df_clean['driver_age_band'] = pd.cut(df_clean['driver_age'],
    bins=[17, 21, 25, 30, 45, 60, 76],
    labels=['18-21', '22-25', '26-30', '31-45', '46-60', '61+'])

# Vehicle age bands: same rationale
df_clean['vehicle_age_band'] = pd.cut(df_clean['vehicle_age'],
    bins=[-1, 2, 5, 8, 12, 20],
    labels=['0-2yr', '3-5yr', '6-8yr', '9-12yr', '13+yr'])

# NCD as ordered category
ncd_map = {0: 'NCD0', 20: 'NCD20', 25: 'NCD25',
    35: 'NCD35', 45: 'NCD45', 50: 'NCD50'}
df_clean['ncd_cat'] = df_clean['ncd_level'].map(ncd_map)
```



```

# Interaction term: SUV in Metro (highest risk combination)
df_clean['suv_metro'] = (
    (df_clean['vehicle_segment'] == 'SUV') &
    (df_clean['region'] == 'Metro')
).astype(int)

# Severity response variable: log(avg_severity) for Gamma GLM
sev_df = df_clean[df_clean['claim_count'] > 0].copy()
sev_df['avg_severity_capped'] = sev_df['claim_cost_capped'] / sev_df['claim_count']

# — 6. ENCODING —————
# One-hot encode: GLM handles categories naturally via dummy vars
# Reference levels: Hatchback (segment), Tier2 (region), NCD0 (ncd), 31-45 (driver_age_band)
cat_features = ['vehicle_segment', 'region', 'ncd_cat', 'driver_age_band',
                'vehicle_age_band', 'channel']

df_encoded = pd.get_dummies(df_clean, columns=cat_features, drop_first=False)
# Drop reference levels manually for clean GLM output
ref_cols = ['vehicle_segment_Hatchback', 'region_Tier2', 'ncd_cat_NCD0',
            'driver_age_band_31-45', 'vehicle_age_band_3-5yr', 'channel_Broker']
df_encoded.drop(columns=[c for c in ref_cols if c in df_encoded.columns], inplace=True)

df_clean.to_csv('motor_od_clean.csv', index=False)
sev_df.to_csv('motor_od_severity.csv', index=False)
print(f"Clean dataset: {df_clean.shape[0]:,} rows, {df_clean.shape[1]} columns")
print(f"Severity dataset: {sev_df.shape[0]:,} rows")

```

07 Modelling Framework

The Two-Part Multiplicative Model

The standard actuarial approach to motor pricing decomposes the **pure premium** (expected claim cost per unit of exposure) into two components:

Pure Premium = Frequency × Severity

WHERE FREQUENCY = E[N/EXPOSURE] AND SEVERITY = E[X | N ≥ 1]

Each component is modelled separately using a Generalised Linear Model (GLM). This separation is important because **the drivers of frequency and severity are often different**. For example, young drivers have higher frequency (more accidents) but not necessarily higher severity (accidents at lower speeds).

COMPONENT	DISTRIBUTION	LINK FUNCTION	OFFSET	RESPONSE VARIABLE
Frequency	Poisson	Log	log(exposure)	Claim count N
Severity	Gamma	Log	None	Avg cost X per claim
Combined	Tweedie (p≈1.5)	Log	log(exposure)	Total claim cost

Why Poisson for Frequency?

- **Claims are rare independent events** — exactly the Poisson assumption.
- The Poisson distribution is a member of the Exponential Dispersion Family (EDF), enabling GLM estimation via IRLS.
- The log link produces **multiplicative relativities** — a natural structure for insurance rating factors (e.g., SUV gets 1.18× the base frequency).
- The **offset term log(exposure)** adjusts for varying policy durations without adding a parameter.
- If overdispersion is present (Var[N] >> E[N]), use **quasi-Poisson** (scale parameter $\phi > 1$) or Negative Binomial GLM.

Why Gamma for Severity?

- **Claim costs are strictly positive** — Gamma is defined on $(0, \infty)$.
- The Gamma distribution has a **constant coefficient of variation** — variance grows with mean^2 , matching observed insurance data where high-cost claims have higher absolute variability.
- The log link produces multiplicative factors, consistent with the frequency model.
- The Gamma assumption is tested via residual analysis and Q-Q plots of deviance residuals.

Why Not Tweedie Directly?

MODEL CHOICE RATIONALE

The Tweedie distribution (Poisson-Gamma compound, power parameter $p \approx 1.5$) can model total claim cost directly and handles zero-inflated data naturally. However, the **two-part approach is preferred** because: (1) it provides interpretable diagnostics for each component; (2) frequency and severity models can use different rating factors; (3) it enables separate monitoring of emerging frequency and severity trends — a key management reporting requirement.

GLM Assumptions

1. **Independence:** Each policy is assumed to be an independent risk unit. Multi-policy households violate this — a known limitation.
2. **Mean function:** $E[Y|X] = g^{-1}(X\beta)$ where $g = \log$ link.
3. **Variance function:** $\text{Var}[Y|X] = \phi \cdot V(\mu)$ where $V(\mu) = \mu$ (Poisson) or μ^2 (Gamma).
4. **Linear predictor:** The log of the expected response is linear in the parameters — each factor enters multiplicatively on the original scale.

Strengths and Weaknesses

ASPECT	STRENGTHS	WEAKNESSES
Interpretability	Relativities are directly usable as tariff factors; regulator-	Cannot capture complex non-linear interactions without engineering